

L9b: Pricing American Style Options Contracts using the Cox, Ross, Rubinstein (CRR) Binomial Model

CHEME 5660 Cornell University

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Objectives for Today

In L9a, we priced European options, which can be exercised only at expiration. Standard options on stocks and ETFs are American-style and can be exercised at any time. What is the right to exercise early worth?

Today's concepts

- Distinguish between European and American options: exercise only at expiration, or at any time before expiration.
- Apply the binomial model to price American options: build CRR trees and compute premiums with the decision rule for early exercise. Estimate the same premium by Monte Carlo simulation.
- Decompose option premiums into components: intrinsic value (immediate exercise value) and extrinsic value (time value).

Lectures and Examples

Lecture:
CHEME-5660-L9b-Lecture-American-Derivatives-Intrinsic-Extrinsic-Value-Fall-2026.ipynb

Example 1:
CHEME-5660-L9b-Example-LSM-Premium-Fall-2026.ipynb

Example 2:
CHEME-5660-L9b-Example-AmericanContractsPremium-Fall-2026.ipynb

Example 3:
CHEME-5660-L9b-Example-ExtrinsicValue-Fall-2026.ipynb

Review on your own:
CHEME-5660-L9b-Example-Hull-American-Put-Fall-2026.ipynb

Concept Review: CRR Binomial Pricing Model

American style options contracts allow the holder to exercise the option at any time before or on the expiration date. In L9a, we wrote the premium of an American-style contract as:

$$\mathcal{P}^{\mathsf{AM}} = \sup_{\tau \leq T} \mathbb{E}_{\mathbb{Q}}\Big(\mathcal{D}_{\tau,0}^{-1}(g_y) \cdot V(K, S(\tau))\Big) \geq \mathcal{P}^{\mathsf{EU}}$$

The exercise time τ is chosen using only the information available at that time (a stopping time). The American premium is at least the European one because the holder can choose when to exercise.

The standard method to price American options is the Cox, Ross, Rubinstein (CRR) binomial model.

Reference: Cox, J. C., Ross, S. A., and Rubinstein, M. (1979). Option pricing: A simplified approach. Journal of Financial Economics 7(3): 229.

Concept Review: The CRR Lattice

At each time step, the share price moves up by a factor u with the risk-neutral probability q, or down by a factor d with probability $(1-q)$:

$$q = \frac{\mathcal{D}_{1,0}\left(g_y\right) - d}{u - d}$$

$$u = e^{\sigma\sqrt{\Delta t}}$$

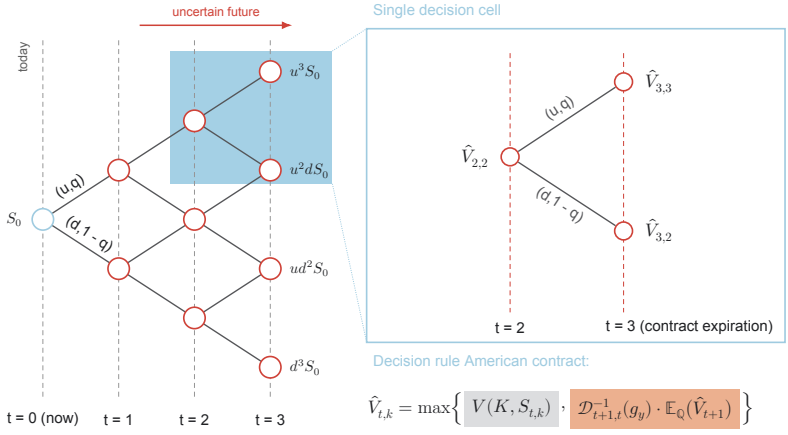
$$d = e^{-\sigma\sqrt{\Delta t}}$$

- Discount factor: $\mathcal{D}_{1,0}\left(g_y\right) = e^{g_y \Delta t}$ for one time step at the risk-free rate g_y.
- L3b's risk-neutral probability: $q = (R_f - d)/(u - d)$ with $R_f = \mathcal{D}_{1,0}(g_y)$.
- Up and down factors: set by the implied volatility σ of the underlying asset and the time step size Δt.

Where do u and d come from? advanced/crr_factors/CHEME-5660-L9b-Advanced-CRR-Factors-Fall-2026.ipynb

Decision-Rule for American Options

At each node, the American holder decides to exercise now or hold for the future.



The Early-Exercise Premium

At expiration, each node's value is the contract's payoff. Working backward, the American holder takes the larger of exercising now and holding for the future, while the European holder can only hold:

$$\hat{V}^{\mathsf{AM}}_{t,k} = \max\Big\{\underbrace{V(K, S_{t,k})}_{\text{exercise now}},\ \underbrace{\mathcal{D}^{-1}_{t+1,t}(g_y)\cdot\left(q\,\hat{V}^{\mathsf{AM}}_{t+1,k+1} + (1-q)\,\hat{V}^{\mathsf{AM}}_{t+1,k}\right)}_{\text{hold for the future}}\Big\}$$

$$\hat{V}^{\mathsf{EU}}_{t,k} = \mathcal{D}^{-1}_{t+1,t}(g_y)\cdot\left(q\,\hat{V}^{\mathsf{EU}}_{t+1,k+1} + (1-q)\,\hat{V}^{\mathsf{EU}}_{t+1,k}\right)$$

Here $\hat{V}_{t,k}$ is the contract's value at the node with k up moves in t steps, and $S_{t,k} = S_0 u^k d^{t-k}$ is the share price there (L3b). The premium is the value at the root, $\mathcal{P} = \hat{V}_{0,0}$. The difference $\mathcal{P}^{\mathsf{AM}} - \mathcal{P}^{\mathsf{EU}} \geq 0$ is the value of the right to exercise early, the early-exercise premium.

A Hypothetical Example from Hull

Consider a put contract.

- Contract: the strike price is $K = 52$ USD/share on a stock whose current price is $S_0 = 50$ USD/share.
- Lattice: two time steps of 1 year each. In each period, the stock price moves up by $u = 1.2$ or down by $d = 0.8$.
- Pricing inputs: the risk-neutral probability of an up move is $q = 0.6282$, and the continuously compounded risk-free rate is $g_y = 5\%$.

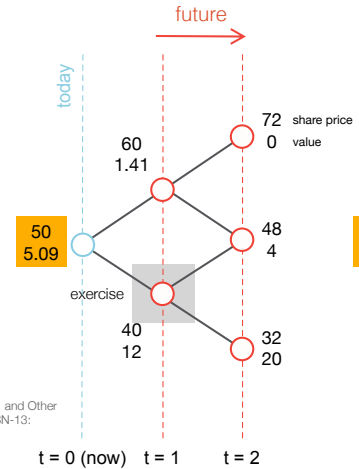
What is the price of the American and European put option contracts?

Source: Chapter 13 of Hull, Options, Futures, and Other Derivatives, 9th edition (2015).

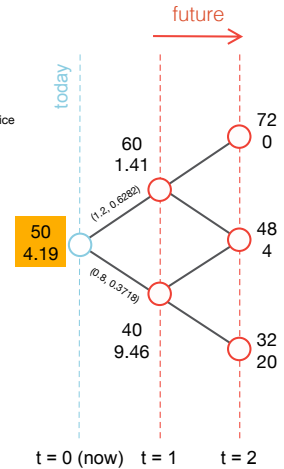
American versus European Put on One Lattice

Example: Hull 13.4 and 13.5

American Put contract



European Put contract



Reference: John C. Hull "Options, Futures, and Other Derivatives" 9th edition, 2015. Pearson, ISBN-13: 978-0133456318

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Early Exercise in the Hull Example

At the down node at $t = 1$, exercising pays 12 USD/share, while holding is worth 9.46 USD/share, so the American holder exercises.

That one decision raises the premium from 4.19 to 5.09 USD/share, an early-exercise premium of about 0.90 USD/share.

Review on your own: CHEME-5660-L9b-Example-Hull-American-Put-Fall-2026.ipynb

Monte Carlo Pricing of American Options

In L9a, we priced European contracts by simulating share prices under $\mathbb{Q}$. Longstaff and Schwartz extended this approach to American contracts. The decision rule needs the expected value of holding, which they estimate by regression, working backward from expiration:

- Simulate: M share price paths under $\mathbb{Q}$ on the exercise dates. At expiration, each path's cash flow is its payoff.
- Regress: the discounted later cash flows of the in-the-money paths on their share prices, giving the continuation value $\hat{C}(S) = \beta_0 + \beta_1 S + \beta_2 S^2$.
- Decide: exercise if $V(K, S(t)) > \hat{C}(S(t))$. The payoff now replaces that path's later cash flow.

Reference: Longstaff, F. A., and Schwartz, E. S. (2001). Valuing American options by simulation: a simple least-squares approach. Review of Financial Studies 14(1): 113–147.

The Early-Exercise Premium by Simulation

The premium is the average discounted cash flow over the paths.

Holding on every path gives L9a's European Monte Carlo premium, so the difference estimates the early-exercise premium.

Example 1: CHEME-5660-L9b-Example-LSM-Premium-Fall-2026.ipynb

Compute the Price of an American Call and Put Option Contract

Now that we've explored the decision rule through the Hull example, let's apply these concepts to real market data.

- Model versus market: CRR prices for American calls and puts from the course's options archive, compared with market prices across strike.
- Calls: on a share that pays no dividend, exercising early never helps. Exercising early means paying the strike sooner and owning the shares if the price then falls below the strike. Holding is always worth at least as much, so $\mathcal{P}_c^{\text{AM}} = \mathcal{P}_c^{\text{EU}}$.
- Puts: exercising early can pay when the put is deep in the money. The holder receives the strike now and earns the risk-free rate on it, so $\mathcal{P}_p^{\text{AM}} > \mathcal{P}_p^{\text{EU}}$ is possible.

Example 2: CHEME-5660-L9b-Example-AmericanContractsPremium-Fall-2026.ipynb

Extrinsic versus Intrinsic Value of American Options

The premium (price) of an American option can be broken down into two components: intrinsic value and extrinsic value.

- Intrinsic value: the immediate exercise value, $\max(0, S - K)$ for a call and $\max(0, K - S)$ for a put, where S is the current stock price.
- Extrinsic value: also known as time value, the additional value from the potential for future price movements before expiration.
- Time decay: as time moves towards expiration, the extrinsic value tends to decrease, because there is less time for the share price to move favorably.

The extrinsic value $E_V(t)$ can be expressed mathematically as:

$$E_V(t) = \text{Option Price} - \text{Intrinsic Value} \quad \blacksquare$$

Extrinsic Value

In expanded form, the extrinsic value $E_V(t)$ is:

$$E_V(t) = \begin{cases} \mathcal{P}_c(K, S(t)) - \max(0, S(t) - K) & \text{(for Call Options)} \\ \mathcal{P}_p(K, S(t)) - \max(0, K - S(t)) & \text{(for Put Options)} \end{cases}$$

When the option is out of the money, it has no intrinsic value, and the entire option price is extrinsic value.

Why do people pay money for something that is "worthless"? There is still a chance that the share price moves favorably before expiration.

Example 3: CHEME-5660-L9b-Example-ExtrinsicValue-Fall-2026.ipynb

Optional Advanced Material

CRR factors: advanced/crr_factors/
CHEME-5660-L9b-Advanced-CRR-Factors-Fall-2026.ipynb

Why does the CRR model use $u = e^{\sigma\sqrt{\Delta t}}$ and $d = e^{-\sigma\sqrt{\Delta t}}$? We match the variance of the lattice's one-step log return to the variance of geometric Brownian motion, which sets the size of the up and down moves. We then check the match on the lattice priced with q, and find how small the time step must be to keep q between zero and one.

Notebooks: lectures/week-9/L9b/advanced, also linked from the lecture notebook.

Summary

Today we priced American options with the CRR binomial model and Monte Carlo simulation, and split option prices into intrinsic and extrinsic components.

Key takeaways

- Binomial trees enable American option pricing: at each node, the holder exercises or holds, so backward induction computes the premium. The American premium exceeds the European premium only when early exercise pays, as for deep in-the-money puts.
- Intrinsic and extrinsic value tell different stories: intrinsic value is the immediate exercise value, and extrinsic value is the potential for favorable price movement before expiration.
- Extrinsic value depends on multiple factors: time to expiration, volatility, and moneyness. Longer-dated options typically carry more extrinsic value.